



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: VII semester B.E. ECE A

Course Code & Title: CS8082 & Machine Learning Techniques

Name of the Faculty member (s): P.Venkatesh

Innovative Practice Description

- **Unit / Topic:** IV / Case Based Reasoning- Instance Based Learning
- **Course Outcome:** CO4 : The students will be able to Suggest and apply appropriate machine learning algorithms for instance based learning problems
- **Topic Learning Outcome:** TLO 13: Apply the concept of Case Based Reasoning learning paradigm for instance based learning
- **Activity Chosen:** Jigsaw

- **Justification:**

The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal each person depends on all the others.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

The lessons are divided into subcategories. Students are divided into groups of four or five students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PSO1
CO2	3	3	2	2	1	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PSO1
	3	3	2	2	1	1	1	2
Justification for correlation	Determining the stability of a given system requires strong knowledge in	The students will analyze the complex engineering	This PO is mapped to level 3 because	As the students use MATLAB	During preparation of assignment and report/feedback	The students will function effectively	Presentation of the results and conclusions of the activity	This PSO has been mapped with level

advanced mathematical techniques like construction of Routh tables and communication engineering concepts and hence PO1 is mapped with level 3	problems and arrive the conclusion about stability of the system using mathematical principles.	students will refine the conceptual design of root locus into a detailed design using graphs with the same constraints	to plot the root locus graph on giving the transfer function, PO5 has been mapped	on collaborative activities, students should not indulge themselves in any unethical activities and hence this PO has been mapped	as a team to be involved in group activities and also as an individual and hence this PO is mapped	carried out, by the students, require good communication and presentation skills and so PO 10 is mapped	2 as the methods available in the unit will help the students to solve existing stability oriented problems in control engineering
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• **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students feel that it was interesting. They learn how to communicate with team members and work together.

- ❖ ***Benefit of the practice:***

- Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

- The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ https://www.educationworld.com/a_curr/strategy/strategy036.shtml
<https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>

Unit / Topic: Unit II / Genetic Models of Evaluation and Learning

- **Course Outcome:** CO 5 / Students will be able apply the concept of neural network back propagation algorithm and genetic algorithm for problem solving

- **Topic Learning Outcome:** TLO 6

- **Activity Chosen:** Think Pair Share

- **Justification:**

Think-pair-share (TPS) is a collaborative learning strategy where students work together to solve a problem or answer a question about an assigned reading. This strategy requires students to (1) think individually about a topic or answer to a question; and (2) share ideas with classmates. Discussing with a partner maximizes participation, focuses attention and engages students in comprehending the reading material. Hence this topic was chosen

- **Time Allotted for the Activity:** 40 minutes

- **Details of the Implementation:**

The activity was planned for duration of 40 minutes. First the students were asked to think about the topic given. Then they were paired among their nearby students. They should then share the ideas that they have recollected and list it down in a paper. Later some among the students (pair) were asked to come front and share their ideas that they have written with the entire class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO3
CO2	3	3	2	2	2	2	2	3	2
(1 – Low			2 – Moderate			3 – High)			

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
	3	2	2	1	1	1	2
Justification for correlation	Solve problems in topics like controllability require strong knowledge in mathematical topics like matrices and Numerical methods and	The students will analyze the engineering problems like controllability and observability of a given system and perform	PO3 is moderately mapped with level 2 because students will build and transfer models to determine the equivalence between	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at	It helps the students to present the topic of discussion as a team, with smooth integration of contributions from all	The students will present the results and conclusions of the activity carried out in front of the class which require good communication and	The students will solve existing/novel problems in control engineering.

	hence PO1 has been mapped strongly to level 3	computations to solve the given mathematical model and so this PO is mapped moderately with level 2	state equations and transfer functions	level 2	individual efforts. So, it is mapped at level 2	presentation skills and so PO 10 is mapped	
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- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***
 - Students feel that they have improved self-learning.
 - They learn how to communicate with team members and work together.
 - They were completely involved
- **Benefits:**
 - i) The personal interaction motivates students who might not generally be interested in the discipline.
 - ii) It engages the entire class and allows quiet students to answer questions without having to stand out from their classmates.
 - iii) Recollection of ideas will be easier as they are combined as pairs
- **Challenges:**
 - i) One of the biggest challenges of the think-pair-share is to get all students to truly be engaged.
 - ii) That is students among different pairs may share the same ideas. In such case instructor may ensure the genuineness.

References:

- i) M. Morris Mano and Michael D. Ciletti, “Digital Design”, 5th Edition, Pearson, 2014.
- ii) Charles H. Roth, Jr. and Larry L. Kinney “Fundamentals of Logic Design”, 7th Edition, Pearson, 2014.